

# How Does the Model Find the "Best" Line?

● Intermediate

The single most important idea in this entire course.

## THE CORE PROBLEM

There are infinitely many possible lines you could draw through a scatter of points. Which one is "best," and what does "best" even mean mathematically?

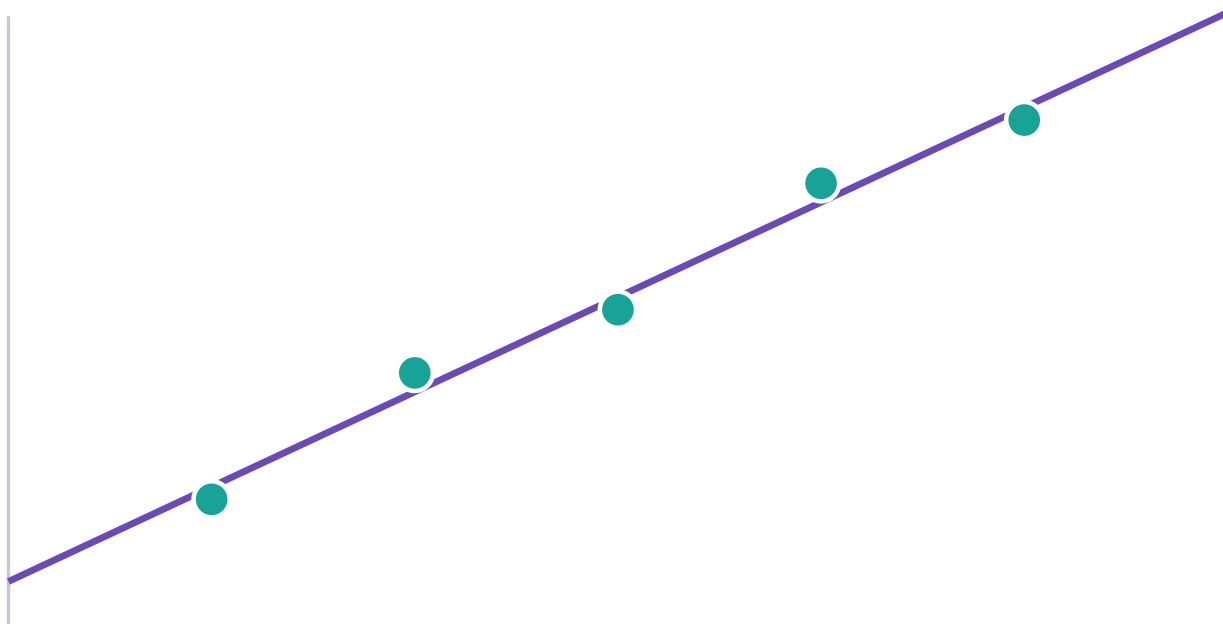
## Predicted value, actual value, error, residual

### DEFINITION

$\text{Residual} = \text{Actual} - \text{Predicted}$

A residual is the vertical gap between a real data point and where the line said it should be. A point above the line has a positive residual; a point below has a negative one.

## Interactive Regression Playground



$$\hat{y} = 0.70 + 1.50x \mid \text{SSE} = 0.30$$

Drag any point (Advertising Spend vs. Sales, from Chapter 7's dataset). The best-fit (OLS) line recalculates live, and SSE updates below.

## Why we don't simply add up raw errors

### TINY EXAMPLE

Actual = 10, Predicted = 8 → Error = 2

Actual = 5, Predicted = 7 → Error = -2

$$2 + (-2) = 0$$

The total looks like a perfect model — zero total error — even though *both* predictions were wrong. Positive and negative errors cancel out and hide real mistakes.

#### WHY SQUARING FIXES THIS

Squaring makes every error non-negative, so errors can no longer cancel:  $(2)^2 = 4$  and  $(-2)^2 = 4$ , giving a total of 8 — which honestly reflects that both predictions missed.

Squaring also penalizes large errors disproportionately more than small ones, which turns out to have very convenient mathematical properties (Chapter 5).